

Design and Implementation of a RISC-V Based SoC

VLSI Design Lab Final Project

Group 07

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Department of Electrical Engineering

Architecture of RISC-V Core and SoC

Core Architecture:

- RV32I ISA (32-bit Integer)
- 5-Stage Pipeline (F-D-E-M-W)

SoC Level Integration:

- AHB/APB Bus Interconnect
- Peripherals: GPIO, UART, Timer

Architecture Diagram

[Insert Top-Level Block Diagram or RTL Schematic here]

Implementation Methodology

- **Hardware Description Language:** Verilog-2001
- **Verification Environment:** SystemVerilog Testbench
- **Target FPGA:** Xilinx Artix-7 (XC7A100T)
- **Target ASIC Node:** TSMC 65nm / SkyWater 130nm
- **Synthesis Tools:** Vivado (FPGA), Cadence Genus (ASIC)
- **Place & Route:** Cadence Innovus (ASIC)

FPGA Timing Simulation Results

Post-Synthesis Timing:

- WNS (Worst Negative Slack):
[Value] ns
- WHS (Worst Hold Slack):
[Value] ns

Post-Implementation Timing:

- WNS: [Value] ns
- WHS: [Value] ns
- **Max Frequency (F_{max}):**
[Value] MHz

Timing Waveforms

*[Insert Vivado Timing Summary
screenshot or Simulation Waveform]*

FPGA Utilization Summary

Resource	Used	Available	Utilization (%)
Slice LUTs	[Value]	63,400	[%]
Slice Registers	[Value]	126,800	[%]
Block RAM (36Kb)	[Value]	135	[%]
DSP48E1	[Value]	240	[%]
I/O Buffers	[Value]	210	[%]

Note: Resource numbers extracted from post-placement utilization report.

Modifications for ASIC Transition

- **Memory Replacement:** Replaced FPGA-specific Block RAMs (BRAM) with Standard-Cell SRAM Macros generated via Memory Compiler.
- **Clock Tree Synthesis (CTS):** Removed FPGA MMCMs. Inserted Integrated Clock Gating (ICG) cells and standard buffers to minimize clock skew.
- **Reset Synchronization:** Implemented dual-flop synchronizers for asynchronous reset de-assertion.
- **I/O Pads:** Instantiated specific I/O Pad cells (Power, Ground, Bidirectional) required for the ASIC pad ring.

ASIC Post-Synthesis & Post-PNR Results

Post-Synthesis (Genus):

- Setup Slack: [Value] ps
- Hold Slack: [Value] ps

Post-PNR (Innovus):

- Setup Slack: [Value] ps
- Hold Slack: [Value] ps
- Core Frequency: [Value] MHz
- Operating Voltage: 1.2 V

ASIC Report: Area, Power, and Timing

Area Summary:

- Standard Cell Area: [Value] μm^2
- Macro Area: [Value] μm^2
- Total Chip Area: [Value] μm^2
- Core Density: [Value] %

Power Summary:

- Dynamic Power: [Value] mW
- Leakage Power: [Value] μW
- Total Power: [Value] mW

GDSII Layout

[Insert your routed ASIC Layout screenshot here]

1. FPGA Hardware Implementation

Demonstrating the RISC-V SoC executing code on the Artix-7 board, showcasing UART communication and GPIO toggling.

Link: [Watch FPGA Demo](#)

2. ASIC Simulation & GDSII Walkthrough

Showcasing the Post-PNR Gate-Level Simulation waveforms and a walkthrough of the final routed layout.

Link: [Watch ASIC Walkthrough](#)

Detailed Contribution Table

Member Name	Responsibility	Key Deliverables
Member 1	Core Architecture & RTL	Datapath, Control Unit, ALU
Member 2	FPGA Implementation	Synthesis, Bitstream, UART
Member 3	ASIC Synthesis	Genus flow, constraints, DFT
Member 4	ASIC PNR	Innovus routing, CTS, GDSII

Thank You

Questions?